Active Learning in First-Semester Calculus: An Open Re-analysis of a Randomized Trial at a Hispanic-Serving Institution

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ABSTRACT

Active learning is widely recommended in college mathematics, but much of the evidence comes from studies whose data cannot be checked. We re-analyze the openly shared data from a pragmatic trial in which 1,019 first-semester calculus students at a large Hispanic-Serving Institution were randomly allocated to active-learning (Modeling Practices in Calculus) or lecture sections across 32 sections and 16 registration time slots (Kramer et al., 2023a). Among the 811 students who remained in their assigned section, active-learning students scored 15.8 points higher on a common 0–100 end-of-term learning measure (95% CI 11.8 to 19.8; Hedges' $g = 0.77$) and were 11.7 percentage points more likely to pass with a C or better (80.3% vs. 68.5%; 95% CI 6.4 to 17.1). The active section outperformed its lecture counterpart on the learning measure in 14 of 16 time slots. Because lecture students were more likely to lack a learning score, we estimate Lee bounds; the learning gain remains at least 11.7 points. Effects are similar across gender, ethnicity, transfer status, and prior preparation. All code and derived data are public so that others can reproduce, challenge, and extend these results.

Keywords: active learning — cooperative learning — calculus — randomized trial — open science — Hispanic-Serving Institutions — replication

1. INTRODUCTION

First-semester calculus is a gateway to nearly every science, engineering, and health degree. Large national studies have documented that many students who begin calculus do not succeed in it, and that a poor calculus experience is associated with leaving STEM majors (Bressoud et al., 2015). Teaching that asks students to work problems in class, often in small groups, has been proposed as one remedy.

Meta-analyses support this view across STEM. Springer et al. (1999) found benefits of small-group learning in undergraduate science and mathematics. Freeman et al. (2014) synthesized 225 studies and found higher examination scores and lower failure rates under active learning, and Theobald et al. (2020) found that active learning narrowed achievement gaps for underrepresented students. Laursen et al. (2014) reported benefits of inquiry-based mathematics courses for both women and men. Yet most primary studies are observational, and few share student-level data. Readers usually cannot check how outcomes were defined, how missing data were handled, or whether results depend on analytic choices.

Kramer et al. (2023a) addressed the first problem with a large pragmatic trial that randomly allocated students to active-learning or lecture calculus sections, and addressed the second by releasing the anonymized student-level data under a CC0 public-domain dedication (Kramer et al., 2023b). This paper uses that release to ask three questions. First, are the reported learning benefits reproducible from the public data with independent code? Second, do they hold for the outcome departments track most closely, passing with a C or better, and are they robust to the trial's differential missingness on the learning measure? Third, is the benefit shared across student groups?

This re-analysis continues the author's undergraduate research on cooperative and active learning in calculus, begun in 2016, with a larger, randomized, and openly available dataset.

We describe the data and trial design in Section 2.1 and our estimation methods in Section 2.3. In Section 3 we verify the reproduced sample against the original documentation. In Section 4 we present effects on learning (Section 4.1), passing (Section 4.2), and student groups (Section 4.3). In Section 5 we discuss these findings, in Section 6 we set out the caveats of this analysis, and in Section 7 we provide the conclusions.

2. METHODS

2.1. Data and Trial Design

The data are the public release accompanying Kramer et al. (2023a), deposited on Dryad and Zenodo (Kramer et al., 2023b). The trial took place over three semesters

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at a large Hispanic-Serving Institution. Students registered for a time slot; within each of the 16 slots, they were randomly allocated to one active-learning section using the Modeling Practices in Calculus curriculum or to one lecture section. In total, 516 students were allocated to active learning and 503 to lecture, across 32 sections taught by 19 instructors (7 active, 12 lecture). No instructor taught in both conditions.

The file contains one row per allocated student with condition, trial status, course grade, an end-of-term learning measure scaled 0–100, a prior-preparation index (the average of SAT/ACT, high school GPA, and placement scores, scaled 0–100), and demographic fields. The reanalysis used the file exactly as released; its SHA-256 fingerprint is recorded in the replication package.

2.2. Analytic Sample and Outcomes

The primary sample is the 811 students recorded as remaining in their assigned section after the drop/add deadline (417 active, 394 lecture). Students who switched to a non-trial section have no outcome in the file. The share remaining did not differ meaningfully by condition (80.8% vs. 78.3%, $p = .325$).

Outcomes were (a) the end-of-term learning measure, available for 376 active and 310 lecture students; (b) passing with a grade of C or better; and (c) a D, F, or withdrawal (DFW) outcome, with withdrawals and other non-credit outcomes counted as DFW.

2.3. Estimation

Treatment effects were estimated by ordinary least squares with time-slot fixed effects, which compare students within the same randomization block,

$$Y_{is} = \tau A_i + \alpha_s + \varepsilon_{is}, \tag{1}$$

where Y_{is} is the outcome of student i in slot s, A_i indicates allocation to active learning, α_s is a slot fixed effect, and τ is the effect of interest. Because outcomes within a section are correlated and instructors are nested in condition, standard errors are cluster-robust by section (CR1; 32 clusters; t reference with 31 degrees of freedom; Cameron & Miller, 2015). A second specification adds the prior-preparation index (mean-centered, with a missing-value indicator). The binary pass outcome is modeled as a linear probability so that τ reads directly in percentage points. Standardized effects on the learning measure are Hedges' g with a pooled standard deviation (Hedges, 1981).

As a design-based check that does not depend on student-level modeling, each slot's active-section mean was compared with its lecture-section mean, giving 16 paired differences. These were tested with a paired t-test and an exact sign-flip randomization test over all 2^{16} sign patterns.

The learning measure was missing for 21.3% of lecture students but only 9.8% of active-learning students. If missingness is related to achievement, simple comparisons may be biased. We therefore computed Lee (2009) bounds, trimming the 12.7% of the active-learning score distribution needed to equalize observation rates, from the top for the lower bound and from the bottom for the upper bound.

Subgroup effects were estimated separately for women and men, Hispanic or Latino/a and other students, Black or African American students, first-time-in-college and transfer students, and terciles of prior preparation. Differences between groups were tested with condition-by-group interaction terms added to Equation 1. With modest subgroup sizes and multiple comparisons, these analyses are exploratory.

All analyses were written in Python independently of the original authors' R code. Every number reported here is generated by the scripts and written to a single statistics file from which this manuscript is built.

3. THE REPRODUCED SAMPLE

The public file reproduced the counts reported by the original authors for allocated students (1,019), students remaining at drop/add (417 active, 394 lecture), sections, slots, and instructors. The lecture count with a learning score matched exactly (310); the active-learning count was 376, three more than the 373 stated in the data documentation. Prior preparation was balanced at allocation (active 65.2, lecture 65.9; $p = .356$).

4. RESULTS

4.1. Learning

Students in active-learning sections averaged 63.5 on the end-of-term learning measure, compared with 47.7 in lecture sections (Table 1; Figure 1). Within time slots, the estimated effect was 15.8 points (95% CI 11.8 to 19.8; $p < .001$), and 16.3 points after adjusting for prior preparation. The standardized effect was $g = 0.77$ (95% CI 0.62 to 0.93).

The section-level analysis gave the same answer: the active section scored higher in 14 of 16 slots, with a mean paired difference of 16.1 points (exact randomization $p < .001$; Figure 2; Table 2 in Appendix A).

Lee bounds placed the learning effect between 11.7 and 20.7 points (g between 0.57 and 1.02). Even under the assumption most unfavorable to active learning, the benefit remained large. Consistent with selective missingness, lecture students without a score had lower prior preparation (63.3) than lecture students with one (66.7).

4.2. Passing and DFW Rates

80.3% of active-learning students passed with a C or better, compared with 68.5% of lecture students. The within-slot effect was 11.7 percentage points (95% CI 6.4

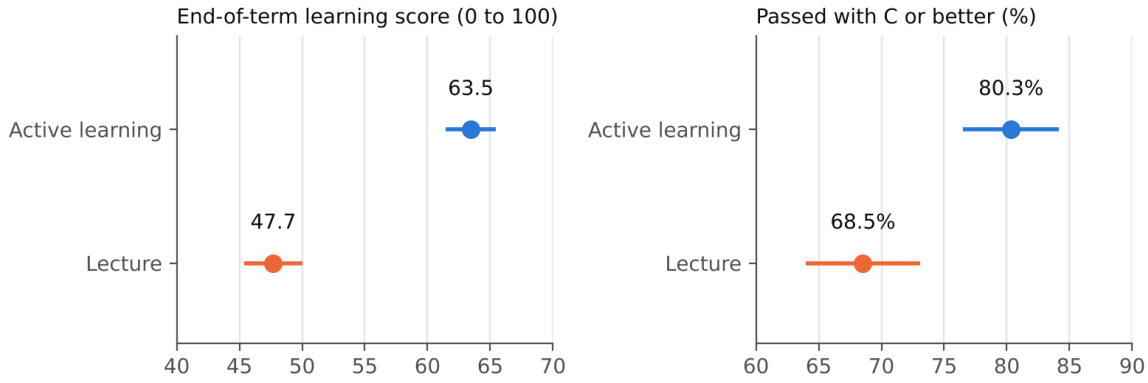


Figure 1. Left: Mean end-of-term learning measure by condition. **Right:** Share passing with a C or better. Students who remained in their assigned section; bars are simple 95% confidence intervals. Clustered estimates are given in Table 1.

TABLE 1. Outcomes and estimated effects of active learning

Outcome	Active learning	Lecture	Estimated effect [95% CI]
Students in assigned section at drop/add	417	394	...
Learning measure, mean (SD)	63.5 (19.8)	47.7 (21.0)	
Time-slot fixed effects			15.8 [11.8, 19.8] points
Adding prior preparation			16.3 [12.8, 19.7] points
Hedges' g			0.77 [0.62, 0.93]
Lee bounds for missing scores			11.7 to 20.7 points
Passed with C or better	80.3%	68.5%	
Time-slot fixed effects			11.7 [6.4, 17.1] pp
Adding prior preparation			11.9 [6.8, 17.0] pp
D, F, or withdrawal (DFW)	19.7%	31.5%	37.5% relative reduction

NOTE—Students who remained in their assigned section ($n = 811$). Effects are from ordinary least squares with time-slot fixed effects and standard errors clustered by section (32 clusters). pp = percentage points.

to 17.1; $p < .001$) and was essentially unchanged by adjustment for prior preparation (11.9 points). Equivalently, the DFW rate fell from 31.5% to 19.7%, a 37.5% relative reduction. At the section level, the active section had the higher pass rate in 12 of 16 slots (exact randomization $p = .008$).

4.3. Student Groups

Gains on the learning measure were similar in every group examined, ranging from 15.4 to 17.3 points (Figure 3; Table 3 in Appendix B). Pass-rate gains were more variable, as expected with smaller samples, and were largest among Hispanic or Latino/a students (14.5 points) and transfer students (16.7 points). Among Black or African American students ($n = 104$), pass rates were 70.0% under active learning and 51.9% under lecture, a difference whose interval narrowly included zero. No condition-by-group interaction was statistically reliable (smallest $p = 0.16$), so the data do not show that any group benefited more or less than others.

5. DISCUSSION

Using only the public data and independent code, this re-analysis reproduced the central finding of Kramer et al. (2023a): randomly allocated students in active-learning calculus sections learned substantially more than those in lecture sections. It adds three things. First, the benefit extends to course passing, the outcome that determines whether students continue in STEM: roughly 12 more students per hundred passed. Second, the learning benefit survives a worst-case treatment of the trial's differential missing data. Third, the benefit appears broadly shared: at an institution where most students are Hispanic or Latino/a, active learning raised outcomes across gender, ethnicity, transfer status, and levels of prior preparation.

The size of the learning effect ($g \approx 0.77$) is larger than the average effects on examination scores in STEM-wide meta-analyses (Freeman et al., 2014). Several features may contribute: a full curriculum redesign rather than added activities, a common learning measure designed for the study, and a comparison with lecture courses at a single institution. These same features limit generalization, which is why open data matter: other institutions

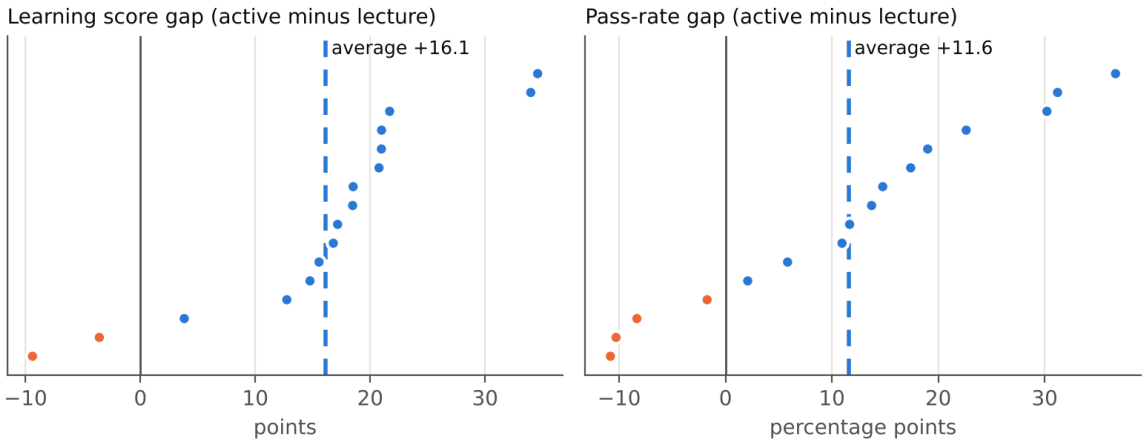


Figure 2. Differences between the active-learning and lecture sections within each of the 16 registration time slots. **Left:** learning measure. **Right:** pass rate. The dashed line marks the mean difference; blue points favor active learning and orange points favor lecture.

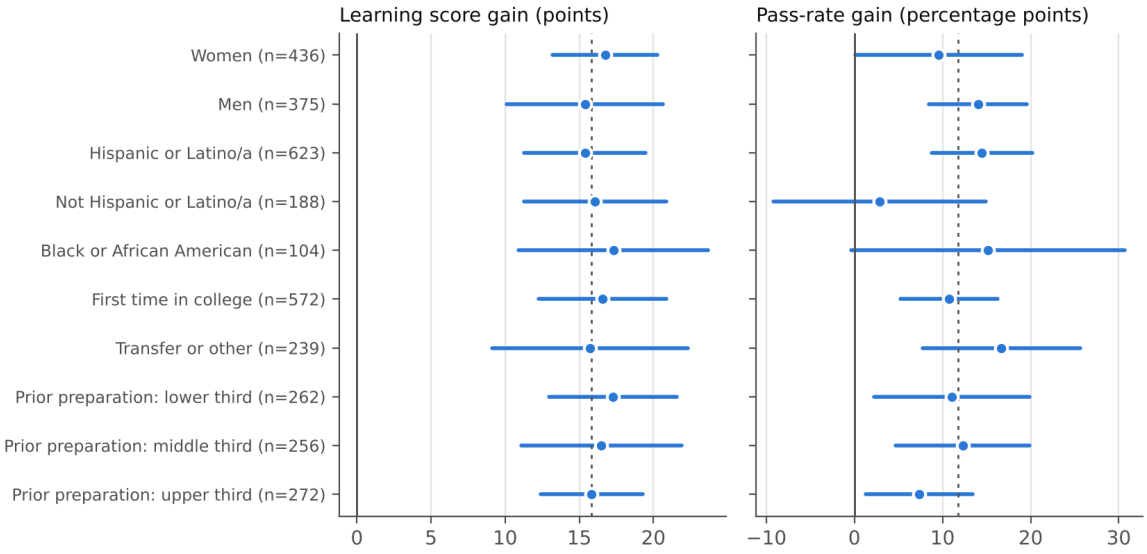


Figure 3. Estimated effects of active learning within student groups (exploratory). **Left:** learning measure. **Right:** pass rate. Lines are 95% confidence intervals clustered by section; the dotted line is the whole-sample estimate.

can apply the same code to their own course records and test whether the effect replicates.

6. CAVEATS

Secondary analyses of a single trial carry their own limitations, and we want to be transparent about them. First, this is one trial at one institution, so results may not generalize to other student populations or curricula. Second, instructors were not crossed with conditions (7 taught active sections and 12 taught lecture), so instructor differences cannot be fully separated from the teaching approach; clustering by section widens intervals to reflect this but does not remove it.

Third, course grades were assigned by instructors who knew the condition, so the pass-rate outcome is less standardized than the common learning measure. Fourth, the learning measure was missing more often in lecture sections; Lee bounds address this under a monotonicity assumption. Fifth, students who switched to non-trial sections have no outcomes, so estimates apply to students who remained in their assigned section. Sixth, subgroup analyses are exploratory. Finally, the data documentation reports three fewer active-learning students with a learning score than the file contains; this does not affect the conclusions.

7. CONCLUSION

We re-analyzed the openly released data from a randomized trial of active learning in first-semester calculus, using independent code and a fully reproducible pipeline. We find four main conclusions.

- Active learning raised end-of-term learning by 15.8 points on a 0–100 scale ($g = 0.77$), reproducing the central result of Kramer et al. (2023a) from the public data alone.
- The benefit extends to course passing: 80.3% of active-learning students earned a C or better, compared with 68.5% of lecture students, and the DFW rate fell by 37.5%.
- The learning gain is robust to differential missing data, remaining at least 11.7 points under worst-case Lee bounds, and appears in 14 of 16 section pairs.
- The benefit was similar across gender, ethnicity, transfer status, and prior preparation; no student group showed a reliably different effect.

Open, checkable evidence of this kind lets departments decide how to teach calculus with confidence. The next version of this project will place these results in national context by extracting the calculus and mathematics studies from published meta-analyses, and will invite departments to run the open pipeline on their own course records.

The author thanks Kramer, Fuller, Watson, Castillo, Duran Oliva, and Potvin for making their trial data openly available. This research received no funding. The author declares no competing interests. This study is a secondary analysis of fully anonymized, publicly released data and involved no contact with participants. Claude (Anthropic) assisted with writing analysis code, building figures, and drafting text; the author directed the analysis, checked every reported number against the data, and takes full responsibility for the content.

Data availability: The original data are available from Dryad and Zenodo under CC0 (Kramer et al., 2023b). All code, derived data, quality-check reports, and figures are available at `https://github.com/faithesan-hub/calculus-active-learning` and archived on Zenodo; the DOI is listed in the repository README.

Software: pandas (McKinney, 2010), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Matplotlib (Hunter, 2007).

APPENDIX

A. SECTION-LEVEL COMPARISONS

Table 2 lists the learning-measure and pass-rate comparisons for each of the 16 registration time slots that form the design-based check in Section 4.1. Each slot contained one active-learning section and one lecture section, so these comparisons hold the time of day and day of week fixed.

TABLE 2. Section-level comparison within each registration time slot

	n with score		Mean learning score			Pass rate	
Slot	Active	Lecture	Active	Lecture	Difference	Active	Lecture
1	28	20	71.3	52.8	+18.5	78.1%	60.7%
2	29	21	54.9	64.3	−9.4	86.2%	71.4%
3	26	22	68.7	47.7	+21.0	66.7%	75.0%
4	17	20	61.7	27.1	+34.6	61.9%	63.6%
5	23	18	51.1	54.7	−3.6	95.8%	90.0%
6	29	22	58.1	45.3	+12.7	93.5%	63.3%
7	17	11	61.6	57.8	+3.8	88.2%	69.2%
8	14	8	53.3	19.3	+34.0	86.7%	50.0%
9	26	25	68.5	53.8	+14.8	60.6%	71.4%
10	30	29	66.2	45.5	+20.8	87.1%	55.9%
11	29	24	71.2	54.0	+17.2	76.7%	65.7%
12	29	21	58.4	39.9	+18.5	81.2%	58.6%
13	34	31	64.7	43.0	+21.7	94.3%	80.6%
14	22	20	74.3	53.3	+21.0	76.0%	73.9%
15	12	9	61.1	44.3	+16.8	86.7%	75.0%
16	11	9	62.0	46.5	+15.5	64.7%	75.0%

NOTE—Each slot contained one active-learning section and one lecture section. Score columns use students with a learning measure; pass columns use all students who remained in the section.

B. STUDENT-GROUP ESTIMATES

Table 3 gives the numerical estimates plotted in Figure 3. All estimates are exploratory and should be read together with the interaction tests in Section 4.3.

TABLE 3. Effects of active learning within student groups (exploratory)

Group	n	Learning gain points [95% CI]	Pass rate, active	Pass rate, lecture	Pass-rate gain, pp [95% CI]
Women	436	16.7 [13.2, 20.3]	79.0%	69.0%	9.5 [0.1, 19.0]
Men	375	15.4 [10.1, 20.7]	82.1%	68.1%	14.0 [8.5, 19.6]
Hispanic or Latino/a	623	15.4 [11.3, 19.5]	82.6%	68.2%	14.5 [8.7, 20.2]
Not Hispanic or Latino/a	188	16.1 [11.3, 20.9]	72.9%	69.6%	2.9 [−9.2, 14.9]
Black or African American	104	17.3 [10.9, 23.7]	70.0%	51.9%	15.2 [−0.3, 30.7]
First time in college	572	16.6 [12.3, 20.9]	83.5%	72.6%	10.7 [5.2, 16.3]
Transfer or other	239	15.7 [9.1, 22.3]	73.0%	58.4%	16.7 [7.7, 25.6]
Prior preparation, lower third	262	17.3 [13.0, 21.6]	68.9%	56.7%	11.1 [2.2, 19.9]
Prior preparation, middle third	256	16.5 [11.1, 21.9]	81.3%	70.1%	12.3 [4.7, 19.9]
Prior preparation, upper third	272	15.8 [12.4, 19.3]	90.4%	82.5%	7.3 [1.3, 13.4]

NOTE—Estimates from ordinary least squares with time-slot fixed effects within each group; confidence intervals clustered by section. Groups overlap and are not mutually exclusive.

REFERENCES

Bressoud, D., Mesa, V., & Rasmussen, C., eds. 2015, Insights and Recommendations from the MAA National Study of College Calculus (Washington, DC: MAA Press)

Cameron, A. C., & Miller, D. L. 2015, Journal of Human Resources, 50, 317, doi: 10.3368/jhr.50.2.317

Freeman, S., Eddy, S. L., McDonough, M., et al. 2014, Proceedings of the National Academy of Sciences, 111, 8410, doi: 10.1073/pnas.1319030111

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Hedges, L. V. 1981, Journal of Educational Statistics, 6, 107, doi: 10.3102/10769986006002107

Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: 10.1109/MCSE.2007.55

Kramer, L., Fuller, E., Watson, C., et al. 2023a, Science, doi: 10.1126/science.ade9803

Kramer, L., Fuller, E., Watson, C., et al. 2023b, Establishing a New Standard of Care for Calculus Using Trials with Randomized Student Allocation [Dataset], Dryad, doi: 10.5061/dryad.kkwh70s95

Laursen, S. L., Hassi, M.-L., Kogan, M., & Weston, T. J. 2014, Journal for Research in Mathematics Education, 45, 406, doi: 10.5951/jresematheduc.45.4.0406

Lee, D. S. 2009, Review of Economic Studies, 76, 1071, doi: 10.1111/j.1467-937X.2009.00536.x

McKinney, W. 2010, in Proceedings of the 9th Python in Science Conference, 56, doi: 10.25080/Majora-92bf1922-00a

Springer, L., Stanne, M. E., & Donovan, S. S. 1999, Review of Educational Research, 69, 21, doi: 10.3102/00346543069001021

Theobald, E. J., Hill, M. J., Tran, E., et al. 2020, Proceedings of the National Academy of Sciences, 117, 6476, doi: 10.1073/pnas.1916903117

Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature Methods, 17, 261, doi: 10.1038/s41592-019-0686-2